

Hyeong-Bin Jeong

**M.S. Student in Mechanical Engineering
at Pusan National University
& Pusan Clean Energy Research Institute**

Combustion & Pyrolysis Research

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Education

- Pusan National University
M.S. in Mechanical Engineering · Major in Energy Systems Engineering
Mar. 2025 – Feb. 2027 (Expected)
- Pusan National University
B.S. in Mechanical Engineering
Mar. 2019 – Feb. 2025

Research Experience

- Energy Conversion Systems Laboratory, PCERI, Pusan National University
M.S. Student Researcher | Mar. 2025 – Present
 - Experimental investigation of COG/NH₃ and H₂/NH₃ co-combustion
 - Flame visualization and exhaust gas analysis under varying fuel and oxidizer conditions
 - 1-D chemical kinetic analysis of flame structure and NO_x formation pathways
- Energy Conversion Systems Laboratory, PCERI, Pusan National University
Undergraduate Research Intern | Mar. 2024 – Feb. 2025
 - Conducted high-pressure pyrolysis experiments of waste plastics
 - Analyzed product gases using gas chromatography (GC)
 - Investigated pressure and catalyst effects on hydrogen production

Publications

- P3

Experimental and Numerical Investigation of Combustion Characteristics and NO Formation in Oxygen-Enriched COG/NH₃ Jet Flames

Hyeong-Bin Jeong, Min-Jong Ku, Kyeong-Ho Kim, Hyeong-Bin Moon, Seung-Mo Kim, Byoung-Hwa Lee, Chung-Hwan Jeon

International Journal of Hydrogen Energy · Under Review · Submitted Jul. 31, 2026

- P2

A strategy for low-carbon steelmaking: Combustion characteristics analysis of steel off-gases and ammonia co-firing

Kyeong-Ho Kim, Min-Jong Ku, **Hyeong-Bin Jeong**, Joonwoo Kweon, Byoung-Hwa Lee, Chung-Hwan Jeon

Fuel, Vol. 428, Part E, Article 140573, 2027 · **Paper** ↗

- P1

Enhanced hydrogen production from polypropylene via NiO/Zeolite Y catalyzed high-pressure pyrolysis: Effects of pressure and methanation reactions

Hyeong-Bin Moon, **Hyeong-Bin Jeong**, Jae-Hun Yang, Ajayan Vinu, Byoung-Hwa Lee, Chung-Hwan Jeon

Journal of the Energy Institute, Vol. 121, Article 102162, 2025 · **Paper** ↗

Conference Presentations

- C3

Effect of Hydrogen Addition on Flame Structure and NO_x Emission Characteristics in Laminar Ammonia Diffusion Flames

Korean Combustion Society Conference · May 2026 · **Oral Presentation**

- C2

Experimental and Numerical Study on the Effect of COG/NH₃ Co-Combustion in a Laminar Diffusion Burner

Korean Combustion Society Conference · Nov. 2025 · **Oral Presentation**

- C1

High-Pressure Catalytic Pyrolysis of Polypropylene: Enhanced Hydrogen Production

Korean Energy & Climate Change Conference · May 2025 · **Oral Presentation**

International Research Experience

- **Osaka University, Japan**

Combustion Engineering Laboratory

· *Visiting Research Student* | **Jan. 2026**

- Conducted experimental investigation of H_2/NH_3 counterflow diffusion flames
- Analyzed flame structure and radical characteristics

- **University of Newcastle, Australia**

- *Global Innovative Centre for Advanced Nanomaterials (GICAN)*

· *Visiting Research Student* | **Jul.–Aug. 2025**

- Conducted catalyst characterization for waste-plastic pyrolysis research
- Performed material characterization using BET, XRD, and SEM-EDS

Technical Skills

- **Experimental & Diagnostics**

FT-IR · Gas Chromatography (GC) · High-Pressure TGA (HP-TGA) · High-Speed / UV Cameras

- **Material Characterization**

XRD · SEM-EDS · BET

- **Numerical & Simulation**

CHEMKIN-PRO — 1-D Flame Simulation, Reaction Pathway & Sensitivity Analysis

ANSYS Fluent — Reactive Flow & Combustion Simulation

- **Engineering & Design**

PTC Creo — 3-D Mechanical Modeling & Component Design